



SAP S/4HANA and SAP S/4HANA Cloud Private Edition

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SAP S/4HANA for behavioral insights

SAP S/4HANA for behavioral insights (abbreviated to: Behavioral Insights) allows organizations, such as government agencies and companies to analyse the behavior of customers and to generate risk scores and insights for multiple scenarios. It provides an expanded overview of events, which enables the organization to perform an efficient and reliable analysis of the customer, so that they can make informed decisions and take appropriate action.

i Note

Behavioral Insights is an Artificial Intelligence (AI) solution and therefore is only available for SAP S/4HANA Cloud Private Edition or SAP S/4HANA on-premise customers with existing licensing.

Quick Reference

This section describes the high-level workflow used in Behavioral Insights.

Installation and Setup

Behavioral Insights is automatically installed within the S/4HANA infrastructure. For more information about the installation process for the S/4HANA infrastructure, see the [Installation Guide for SAP S/4HANA 2020](#).

Enable Behavioral Insights using the Switch Framework

In the S/4 HANA framework, Behavioral Insights is a self-contained business function that you can switch on. From a technical perspective, it is a switch that assigns repository objects to the business function.

You must enable the BF_BEH_INS switch to be able to use the Behavioral Insights product. For more information about enabling the switch, see [Switching on Business Functions](#).

Maintain Configuration Information

Once the Behavioral Insights business function has been enabled using the switch framework, you can customize the implementation information using the Customizing activities.

The Customizing activities for this product are embedded within Customizing for your SAP application. You can use the transaction SPRO to access the Customizing structure.

See the SAP Reference Customizing documentation for information about managing the Customizing activities for the product. The Behavioral Insights Customizing activities are accessed as follows:

SAP Customizing Implementation Guide > Cross-Application Components > Behavioral Insights

Model Scenarios using ISLM

Behavioral Insights uses Intelligent Scenario Lifecycle Management (ISLM) to model and train scenarios. ISLM is a framework that allows you to perform lifecycle management operations on machine learning scenarios. You can view ISLM as a self-service tool that allows you to handle various operation requirements for machine learning scenarios. You can train an intelligent scenario and use the trained model to obtain an accurate inference result.

For detailed information about training prediction scenarios in Behavioral Insights, see [Train Prediction Scenarios using ISLM](#).

Run Prediction Report

In Behavioral Insights, once the prediction scenarios are trained, you can use the Maintain Application Jobs app to schedule and monitor Behavioral Insights-related jobs, using a standard template that is pre-delivered. You can use the app to schedule, monitor or cancel pre-scheduled jobs, which can reduce your workload by running the tasks smoothly in the background. For more information, see [Maintain Application Jobs](#).

You can also run the *BEI_PREDICTION_WRAPPER* prediction report or schedule a job to run the prediction report.

For more information, see [Prediction Report](#).

Perform Analytics using the Explore At-Risk Customers app

You use the Explore At-Risk Customers app to analyze risk scores and trends across the scenarios delivered by Behavioral Insights.

For more information, see the [Explore At-Risk Customer](#) section.

Perform Analytics using the At-Risk Customers Overview app

You can use the At-Risk Customers Overview app to view trends such as collection strategy, customer groups, and more, represented visually, across the multiple machine learning scenarios offered - Receivership, Late Payment, and Late Filing.

For more information, see the [At-Risk Customers Overview](#) section.

Perform Analytics using SAP Analytics Cloud

Once the prediction scenarios are trained and the report is run, the scenario results can be accessed using SAP Analytics Cloud (SAC). You can use SAC to perform analytics on the trained results. For more information, see [Help Library - SAP Help Portal for SAC](#).

For detailed information about performing analytics using SAC for Behavioral Insight, see [Analytics Using SAP Analytics Cloud](#).

Integration with SAP Marketing Cloud

The risk score values for the corresponding customers in each Behavioral Insights scenario can be exported to and updated in SAP Marketing Cloud. This information can then be used for segmentation and target group building in SAP Marketing Cloud.

For information about setting up Behavioral Insights integration with SAP Marketing Cloud, see [Integration with SAP Marketing Cloud](#).

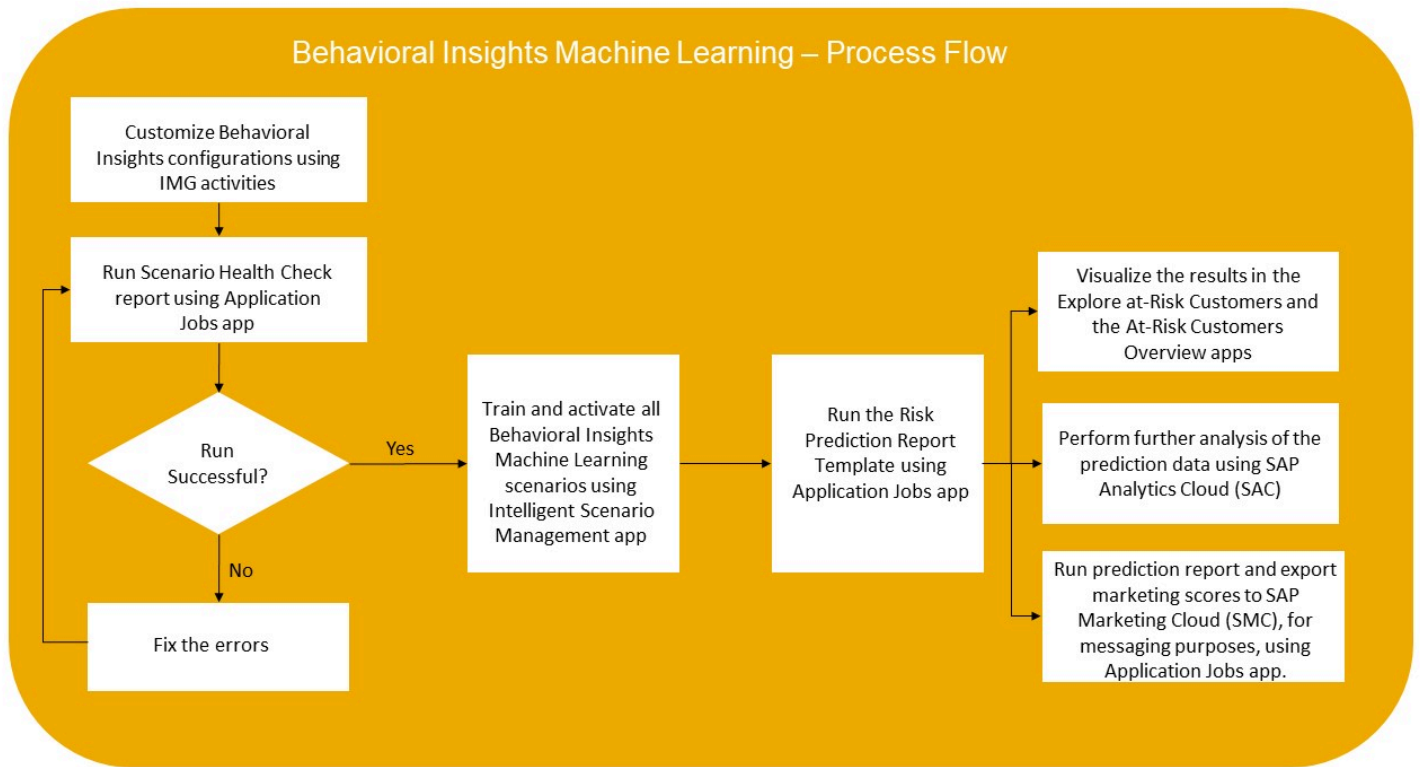
APIs for Behavioral Insights

Behavioral Insights offers the External Events - Read, Create (A2X) API that enables you to read or create Behavioral Insights external event data with the data provided in a payload, in an API call. It is based on the OData protocol, and can be consumed in SAP Fiori apps and on other user interfaces.

For information about this API, see [APIs for Behavioral Insights](#).

Behavioral Insights Machine Learning – Process Flow

The diagram below illustrates the end-to-end process flow for the machine learning scenarios:



Key Terms

Event

An event is a historical record of an interaction, transaction, or activity that occurs for a customer, account, or sub-account. Some examples of events are tax filing, late payment, late submission and so on.

Scenario

A scenario is a use case for which the prediction is to be made. Each scenario can be defined to make predictions at customer, account, or subaccount level.

Account

An account can represent a product or service line. A customer can have multiple accounts. For example, a customer can maintain an account for each tax type.

Subaccount

A subaccount is found at the level below an account and can represent a further classification, such as the location of the customer account. A customer or an account can have multiple subaccounts.

Reason Code

A reason code is a contributing factor or a feature to explain the risk score for a scenario. Some examples of reason codes are the due amount in the local currency, business partner type, and so on.

Prediction Value

The value predicted based on the context of the scenario.

Debt

The amount that an entity owes. The amount is calculated at customer, account or subaccount level for the different scenarios.

Debt Range

The debt is categorized as high, medium and low according to the debt range.

i Note

The debt range must be configured by your administrator.

Customer Status

The customer status from the most recent "customer status change" event. For example, live bankruptcy can be a customer status event.

ISLM Machine Learning Scenarios

Behavioral Insights uses the S/4HANA framework called Intelligent Scenario Lifecycle Management (ISLM) to train and predict machine learning outcomes. The default scenarios that are delivered using the *BEI_ML* package are:

- *BEI_RECEIVERSHIP*: Receivership
- *BEI_LATE_PAYMENT*: Late Payment
- *BEI_LATE_FILING*: Late Filing
- *BEI_LATE_PAYMT_CLSTR*: Customer Groups

Receivership

Business Use Case

The Receivership scenario includes content to predict the entities that have a risk of going into receivership. This is done by calculating the probabilities for each of the entities based on different features that each entity has performed.

Who

Companies who pay a specific tax.

Data Source

Pre-defined source event data from SAP S/4HANA or external event sources.

Output

- Risk score (0-100) that customer may go into receivership (or other user-defined customer status).
- Top reason codes for receivership.

Business Benefits

- Maximize collections and success by focusing on customers with the highest risk first.
- Reduce collection costs by intervening sooner and understanding the unique nature of each customer's journey.
- Transform your processes by quickly deploying machine learning scenarios for customer and behavioral insights.

Reason Codes for Receivership

SAP provides a list of pre-defined reason codes, which are sometimes referred to as features or contributing factors, which are used in the receivership scenario.

Each feature is calculated for a defined interval, such as for the past 2, 6, or 12 months, before each status change for each entity.

Reason Code	Reason Text
BusinessPartnerType	Business Partner Type
BPGroupingCharacter	Business Partner Segment
NrOfLatePaymentPenalty12Month	Payment Penalties (Past 12 mos)
NrOfLateSubmsnPenalty12Month	Submission Penalties (Past 12 mos)
NumberOfLateSubmission12Month	Late Submissions (Past 12 mos)
NumberOfTaxCorrection12Month	Tax Corrections (Past 12 mos)
NrOfInterestDocument12Month	Interest Documents (Past 12 mos)
NrOfNoClearingDocument12Month	No Clearing Documents (Past 12 mos)
NumberOfTransferPosting12Month	Transfer Postings (Past 12 mos)
NumberOfCreatedAudit12Month	Audits Created (Past 12 mos)
NumberOfAuditAssessment12Month	Audit Assessments (Past 12 mos)
NumberOfObjection12Month	Audit Objections (Past 12 mos)
NumberOfNoSubmission12Month	Missed Submissions (Past 12 mos)
NrOfNoClearingDocument6Month	No Clearing Documents (Past 6 mos)
NrOfLatePaymentPenalty6Month	Payment Penalties (Past 6 mos)
NumberOfLateSubmission6Month	Late Submissions (Past 6 mos)
NrOfLateSubmsnPenalty6Month	Submission Penalties (Past 6 mos)
NrOfLatePaymentPenalty2Month	Payment Penalties (Past 2 mos)
NumberOfLateSubmission2Month	Late Submissions (Past 2 mos)
NrOfLateSubmsnPenalty2Month	Submission Penalties (Past 2 mos)
NrOfNoClearingDocument2Month	No Clearing Documents (Past 2 mos)
NumberOfLevel01DunningNonFiler	Non-Filer Dunning L1 (Past 12 mos)

Reason Code	Reason Text
NumberOfLevel02DunningNonFiler	Non-Filer Dunning L2 (Past 12 mos)
NumberOfLevel03DunningNonFiler	Non-Filer Dunning L3 (Past 12 mos)
NumberOfLevel04DunningNonFiler	Non-Filer Dunning L4 (Past 12 mos)
NumberOfLevel01DunningNonPayer	Non-Payer Dunning L1 (Past 12 mos)
NumberOfLevel02DunningNonPayer	Non-Payer Dunning L2 (Past 12 mos)
NumberOfLevel03DunningNonPayer	Non-Payer Dunning L3 (Past 12 mos)
NumberOfLevel04DunningNonPayer	Non-Payer Dunning L4 (Past 12 mos)
NumberOfDunning6Month	Dunning Steps (Past 6 mos)
NumberOfDunning2Month	Dunning Steps (Past 2 mos)
DueAmountInLocalCurrency	Due Amount (MUR)
ChgInDueAmtValue	Average Open Amount Changed (MUR)
ExternalEventHasAffected	Affected External Events (Past 3 mos)

Late Payment

Business Use Case

The Late Payment scenario includes content to predict which entities might not pay the amount due by the due date.

Who

Companies who pay a customer-defined tax at subaccount level.

Data Source

Pre-defined source event data from SAP S/4HANA or external event sources.

Output

- Risk score (0-100) that a customer might not pay the amount due by the due date.
- Top reason codes for late payment.

Business Benefits

- Maximize collections and success by focusing on customers with the highest risk first.
- Reduce collection costs by intervening sooner and understanding the unique nature of each customer's journey.
- Transform your processes by quickly deploying machine learning scenarios for customer and behavioral insights.

Reason Codes for Late Payment

SAP provides a list of pre-defined reason codes, which are sometimes referred to as features or contributing factors, which are used in the late payment scenario.

Each of the features below is calculated on a monthly basis for each entity in a scenario. For example, if the prediction is generated in June 2021, each of the features below is calculated using an interval of 3, 6, or 12 months before June 2021.

Reason Code	Reason Text
BusinessPartnerType	Business Partner Type
BPGroupingCharacter	Business Partner Segment
NrOfLatePaymentPenalty3Month	Payment Penalties (Past 3 mos)
NrOfLateSubmsnPenalty3Month	Payment Submissions (Past 3 mos)
NrOfCreatedAudit3Month	Audits Created (Past 3 mos)
NrOfAuditAssessment3Month	Audit Assessments (Past 3 mos)
NrOfObjection3Month	Audit Objections (Past 3 mos)
NrOfLateSubmission3Month	Late Submissions (Past 3 mos)
NrOfNoSubmission3Month	Missed Submissions (Past 3 mos)
NumberOfLateSubmission6Month	Late Submissions (Past 6 mos)
NrOfNoSubmission6Month	Missed Submissions (Past 6 mos)
NumberOfLateSubmission12Month	Late Submissions (Past 12 mos)
NumberOfNoSubmission12Month	Missed Submissions (Past 12 mos)
NumberOfLevel01DunningNonFiler	Non-Filer Dunning L1 (Past 12 mos)
NumberOfLevel02DunningNonFiler	Non-Filer Dunning L2 (Past 12 mos)
NumberOfLevel03DunningNonFiler	Non-Filer Dunning L3 (Past 12 mos)
NumberOfLevel04DunningNonFiler	Non-Filer Dunning L4 (Past 12 mos)
NumberOfLevel01DunningNonPayer	Non-Payer Dunning L1 (Past 12 mos)
NumberOfLevel02DunningNonPayer	Non-Payer Dunning L2 (Past 12 mos)
NumberOfLevel03DunningNonPayer	Non-Payer Dunning L3 (Past 12 mos)
NumberOfLevel04DunningNonPayer	Non-Payer Dunning L4 (Past 12 mos)
DueAmountInLocalCurrency	Due Amount (MUR)
NrOfMonthWithAuthority	Months Since Registration
NumberOfBankruptcies	Number of Bankruptcies
FilgAmtChgCurAndPrevMonthValue	Difference Amount - Current and Past Month Filing (MUR)
FilgAmtChgCurAndPrevYearValue	Difference Amount - Current and Past Year Filing (MUR)

Reason Code	Reason Text
NumberOfMonthDebtAmountOverdue	Months Overdue
ScaledDebtAmountValue	Scaled Debt Amount (MUR)
NrOfMonthSincePayer	Months Since On-Time Payment
BusinessPartnerStatusName	Status
DeviationInFilingAmount	Filing Amount Deviation (MUR)
AvgNrOfOverdueDay	Average Late Payment Days (Past 3 mos)
ExternalEventHasAffected	Affected External Events (Past 3 mos)

Late Filing

Business Use Case

The Late Filing scenario includes content to predict the entities that may not file a tax return by the due date.

Who

Companies who pay a user-defined tax at subaccount level.

Data Source

Pre-defined source event data from SAP S/4HANA or external event sources.

Output

- Risk score (0-100) that a customer may not file a tax return by the due date.
- Top reason codes for late filing.

Business Benefits

- Maximize collections and success by focusing on customers with the highest risk first.
- Reduce collection costs by intervening sooner and understanding the unique nature of each customer's journey.
- Transform your processes by quickly deploying machine learning scenarios for customer and behavioral insights.

Reason Codes for Late Filing

SAP provides a list of pre-defined reason codes, sometimes called features or contributing factors, used in the late filing scenario.

Each of the below features are calculated on a monthly basis, for each entity in the scenario. For example, if the prediction is being generated in June 2021, then each of the below features are calculated taking an interval of 3, 6, or 12 months before June 2021.

Reason Code	Reason Text
BusinessPartnerType	Business Partner Type
BPGroupingCharacter	Business Partner Segment
NrOfLatePaymentPenalty3Month	Payment Penalties (Past 3 mos)
NrOfLateSubmsnPenalty3Month	Payment Submissions (Past 3 mos)
NrOfCreatedAudit3Month	Audits Created (Past 3 mos)
NrOfAuditAssessment3Month	Audit Assessments (Past 3 mos)
NrOfObjection3Month	Audit Objections (Past 3 mos)
NrOfInterestDocument3Month	Interest Documents (Past 3 mos)
NrOfInterestDocument6Month	Interest Documents (Past 6 mos)
NrOfInterestDocument12Month	Interest Documents (Past 12 mos)
NrOfNoClearingDocument3Month	No Clearing Documents (Past 3 mos)
NrOfNoClearingDocument6Month	No Clearing Documents (Past 6 mos)
NrOfNoClearingDocument12Month	No Clearing Documents (Past 12 mos)
NumberOfLevel01DunningNonFiler	Non-Filer Dunning L1 (Past 12 mos)
NumberOfLevel02DunningNonFiler	Non-Filer Dunning L2 (Past 12 mos)
NumberOfLevel03DunningNonFiler	Non-Filer Dunning L3 (Past 12 mos)
NumberOfLevel04DunningNonFiler	Non-Filer Dunning L4 (Past 12 mos)
NumberOfLevel01DunningNonPayer	Non-Payer Dunning L1 (Past 12 mos)
NumberOfLevel02DunningNonPayer	Non-Payer Dunning L2 (Past 12 mos)
NumberOfLevel03DunningNonPayer	Non-Payer Dunning L3 (Past 12 mos)
NumberOfLevel04DunningNonPayer	Non-Payer Dunning L4 (Past 12 mos)
DueAmountInLocalCurrency	Due Amount (MUR)
NrOfMonthWithAuthority	Months Since Registration
NumberOfBankruptcies	Number of Bankruptcies
FilgAmtChgCurAndPrevMonthValue	Difference Amount - Current and Past Month Tax Filing (MUR)
FilgAmtChgCurAndPrevYearValue	Difference Amount - Current and Past Year Tax Filing (MUR)
NumberOfMonthDebtAmountOverdue	Months Overdue
ScaledDebtAmountValue	Scaled Debt Amount (MUR)
NrOfMonthSinceFiler	Months Since On-Time Tax Filing
BusinessPartnerStatusName	Status
DeviationInFilingAmount	Tax Filing Amount Deviation from Last Month (MUR)

Reason Code	Reason Text
AvgNrOfOverdueDay	Average Late Tax Filing Days (Past 3 mos)

Customer Groups

Business Use Case

Customer Groups machine learning scenario includes content to group the customers who are at a risk of late payment. Groups are automatically generated using Machine Learning based on taxpayer behaviour.

Who

Companies who pay a specific tax.

Data Source

Pre-defined source event data from SAP S/4HANA or external event sources.

Output

- Group ID or the group that the customer is allocated to

Business Benefits

- Transform your processes by proposing potential next steps for a group of customers.
- Propose a strategic decision for a specific group of customers, based on past behavior.
- Reduce collection costs by intervening sooner and understanding the unique nature of each customer's journey.

Predicted Days to Payment

Business Use Case

The Days-to-Payment scenario includes content to predict the days to tax payment for the entities. This prediction is generated using Machine Learning based on taxpayer behaviour.

Who

Companies who pay a specific tax.

Data Source

Pre-defined source event data from SAP S/4HANA or external event sources.

Output

- Number of days to tax payment

Business Benefits

- Maximize tax collection success by predicting the potential days to payment for the customers.
- Reduce collection costs by intervening sooner and understanding the unique nature of each customer's journey.

Train Prediction Scenarios using ISLM

You can use the Intelligent Scenario Management app in ISLM to perform the following procedures:

- [Training the Prediction Scenario](#)
- [Retraining a Prediction Scenario](#)
- [Checking the Quality of a Prediction Scenario](#)
- [Activating a Prediction Scenario](#)
- [Deactivating a Prediction Scenario](#)
- [Deleting a version of an existing Prediction Scenario](#)

Training a Prediction Scenario

You can use Intelligent Scenario Lifecycle Management (ISLM) to train the default scenarios delivered by Behavioral Insights.

Prerequisites

Only users with relevant access can perform this procedure. For more information on the access roles, see [Access the SAP Fiori apps for ISLM](#).

Procedure

1. Launch the SAP Fiori launchpad.
2. Open the **Intelligent Scenario Management** app.

The **Intelligent Scenario Management** page opens with a list of default published intelligent scenarios.

3. Click the Intelligent Scenario that you want to train. For example, **BEI_LATE_FILING**.

The **Train Model** window opens. You can see the details of the intelligent scenario versions.

4. Select the scenario you want to train and click **Train**.
5. You can customize the training data set further using the filters in the **Training Filters** section.
6. Click **Train**.

Retraining a Prediction Scenario

You can also retrain the prediction scenario when there is a decline in the accuracy of the prediction models. The decline may be determined by either verifying the risk scores against the actual prediction dataset or checking the predictive power information. If the predictive power is less than 50%, then the performance of the model may not be accurate. For more information on checking the quality of the Predictive scenario, see [Machine Learning Model Store](#).

Prerequisites

Only users with relevant access can perform this procedure. For more information on the access roles, see [Access the SAP FIORI apps for ISLM](#).

Procedure

1. Launch the SAP Fiori launchpad.
2. Open the **Intelligent Scenario Management** app.

The **Intelligent Scenario Management** page opens with a list of default published intelligent scenarios.

3. Click the Intelligent Scenario that you want to retrain. For example, **BEI_LATE_FILING**.

The **Train Model** window opens. You can see the details of the intelligent scenario version.

4. Click the arrow button to display the versions of the scenarios.
5. Select the prediction scenario version you want to retrain.
6. Click **Retrain**.

Checking the Quality of a Prediction Scenario

You can check the quality of the prediction scenario and can determine based on the predictive power if the scenario needs to be retrained.

Prerequisites

Only users with relevant access can perform this procedure. For more information on the access roles, see [Access the SAP FIORI apps for ISLM](#).

Procedure

1. Launch the SAP Fiori launchpad.
2. Open the **Intelligent Scenario Management** app.

The **Intelligent Scenario Management** page opens with a list of default published intelligent scenarios.

3. Click the Intelligent Scenario for which you want to view the quality information. For example, **BEI_LATE_FILING**.

The **Train Model** window opens. You can see the details of the intelligent scenario versions.

4. Click the arrow button to display the versions of the scenarios.
5. Select the prediction scenario version for which you want to view the prediction quality details.
6. Click the arrow button.

This displays the Model version report for the prediction scenario with the **Quality**, **Predictive Power**, **Prediction Confidence**, and the **Key Influencer** details.

Activating a Prediction Scenario

You can activate an inactive model of a prediction scenario.

Prerequisites

Only users with relevant access can perform this procedure. For more information on the access roles, see [Access the SAP FIORI apps for ISLM](#).

Procedure

1. Launch the SAP Fiori launchpad.
2. Open the **Intelligent Scenario Management** app.

The **Intelligent Scenario Management** page opens with a list of default published intelligent scenarios.

3. Click the Intelligent Scenario that you want to activate. For example, **BEI_LATE_FILING**.

The **Train Model** window opens. You can see the details of the intelligent scenario versions.

4. Click the arrow button to display the versions of the scenarios.
5. Select the prediction scenario version that you want to activate.
6. Click **Activate**.

Deactivating a Prediction Scenario

You can deactivate an active model of a prediction scenario.

Prerequisites

Only users with relevant access can perform this procedure. For more information on the access roles, see [Access the SAP FIORI apps for ISLM](#).

Procedure

1. Launch the SAP Fiori launchpad.
2. Open the **Intelligent Scenario Management** app.

The **Intelligent Scenario Management** page opens with a list of default published intelligent scenarios.

3. Click the Intelligent Scenario that you want to deactivate. For example, **BEI_LATE_FILING**.

The **Train Model** window opens. You can see the details of the intelligent scenario versions.

4. Click the arrow button to display the versions of the scenarios.
5. Select the prediction scenario version that you want to deactivate.
6. Click **Deactivate**.

Deleting a version of an existing Prediction Scenario

You can delete both active and inactive versions of a prediction scenario.

Prerequisites

Only users with relevant access can perform this procedure. For more information on the access roles, see [Access the SAP FIORI apps for ISLM](#).

Procedure

1. Launch the SAP Fiori launchpad.
2. Open the **Intelligent Scenario Management** app.

The **Intelligent Scenario Management** page opens with a list of default published intelligent scenarios.

3. Click the Intelligent Scenario with the version you want to delete. For example, **BEI_LATE_FILING**.

The **Train Model** window opens. You can see the details of the intelligent scenario versions.

4. Click the arrow button to display the versions of the scenarios.
5. Select the prediction scenario version you want to delete.
6. Click **Delete**.

Adding a Custom Scenario

Context

You can add custom scenarios in Behavioral Insights using the following procedure:

Procedure

1. You must update all necessary configurations for the new scenario in the scenario extraction configuration. You must configure the scenario details in the **Maintain Scenario** activity.

For more information about accessing the customizing information, see **Maintain Configuration Information** in the [Quick Reference](#) section.

The IMG activities for this product are embedded within the IMG of your SAP application. You can use the transaction **SPRO** to access the IMG structure. You can use the **SAP Reference IMG documentation** for information on managing all the IMG activities for the product.

The path for the Behavioral Insights IMG is:

► **SAP Customizing Implementation Guide** ► **Cross-Application Components** ► **Behavioral Insights** ►

2. Once the scenario details are configured, you must add the CDS views for your scenario into the Prediction report using the enhancement spots as shown in the image below:

```

WHEN 3.
  lv_pred_dataset = 'C_LATEFILINGPREDICTDATA'.
  lv_train_dataset = 'C_LATEFILINGTRAINDATA'.
  "Getting the unbalanced data from the training dataset (table function).
  IF lt_cfg[ 1 ]-balance_ratio IS NOT INITIAL.
    SELECT BhvrlInsgrsTgtVal AS target, COUNT(*) AS cnts FROM p_tf_late_filing( p_datasettype = 'T', P_BalanceFlag = ' ' )
    GROUP BY BhvrlInsgrsTgtVal INTO TABLE @lt_unbalanced.
  ENDIF.
ENDCASE.
ELSE.
  ex_w_status = 4.
  RETURN.
ENDIF.

ENHANCEMENT-POINT CUSTOM_SCENARIO SPOTS CUSTOM_SCENARIO .

```

3. Open the **WRITE_TO_PRES** method in the **CL_PREDICT_RISK_SCORE** prediction class, using the **SE24** transaction.
4. The enhancement point in the method can be used to add custom scenarios. You must enhance the class and add the following details in the enhancement spot:

Sample Code

```

IF im_w_scenario_id = 99.
  lv_pred_dataset = <Training view name>.
  lv_train_dataset = <Predict view name>.

```

ENDIF.

Results

The new scenario IDs are now taken into consideration and the values are populated accurately for your scenario.

Extending the Explore-At-Risk Customer app with a Custom Field

You can extend the existing *C_BhvrInsightsCustomerList* CDS view used in the [Explore-At-Risk Customer](#) application, in Behavioral Insights, using the procedure detailed in the [Extending Apps with Custom Fields](#) section of the [SAP - ABAP Programming Model for SAP Fiori](#) documentation.

You must add the below annotations to extend the CDS view with additional fields on the [Smart filter](#) bar and table, in the [Explore-At-Risk Customer](#) application:

Sample Code

```
@UI.LINEITEM: { LABEL: '<PREFERRED LABEL>', POSITION: 45 }
@UI.SELECTIONFIELD: [ { POSITION: 45 } ]
```

Prediction Report

In Behavioral Insights, once the prediction scenarios are trained, you can run the prediction scenarios using one of the following methods:

- Using the Maintain Application Jobs app
- Using the *BEI_PREDICTION_WRAPPER* prediction report in ABAP
- Using job scheduling in ABAP

Maintain Application Jobs

You can use the [Maintain Application Jobs](#) app to configure and schedule jobs to run prediction reports, health check reports for scenario, and also export marketing scores, using the templates that are pre-delivered by Behavioral Insights.

You can use this app to:

- Schedule jobs based on the job template delivered by Behavioral Insights
- Schedule recurring jobs
- Copy and cancel jobs
- Monitor jobs based on their status
- Display job results based on their status

For a more detailed overview of the standard key features offered by this app, see [Application Jobs](#).

Behavioral Insights provides the following job templates:

- **Risk Prediction Report Template for Behavioral Insights:** You can use this template to run the prediction report.
- **Scenario Health Check Report Template for Behavioral Insights:** You can use this template to check the health of the prediction report for each scenario.
- **Marketing Scores Export Report Template for Behavioral Insights:** You can use this template to export the marketing scores for each scenario.
- **Risk Prediction with Marketing Scores Export Report Template for Behavioral Insights:** You can use this template to sequentially run the prediction report and then export the marketing scores, for each scenario.

Running the Prediction Report

Context

You can run the *BEI_PREDICTION_WRAPPER* prediction report using the *SE38* transaction. You can also schedule a job to run the prediction report using ABAP. For more information, see [Job Scheduling Explained](#).

i Note

- Running the prediction report using the *BEI_PREDICTION_WRAPPER* is optional if the **Maintain Application Jobs** app is not used.
- We recommend that you run the prediction report as a scheduled job.

Procedure

1. In ABAP, execute the *SE38* transaction.

The **ABAP Editor: Initial screen** is displayed.

2. Specify *BEI_PREDICTION_WRAPPER* as the name of the program and click **Execute**.
3. Select the scenario ID for which you want to run the prediction report from the list and click **Execute**.

A status message indicates whether the prediction report has run successfully.

Situation Handling in Behavioral Insights

Behavioral Insights offers standard Situation Handling features. When the prediction report is run, based on the situation objects and templates are set up in the background by the administrator for specific situations, events are triggered.

The following high levels steps need to be performed by the administrator to configure Situation Handling in Behavioral Insights:

1. Create Situation Objects
2. Create Situation Scenarios
3. Create Situation Templates
4. Create Situation Type
5. Create Events

For more information on configuring Situation Handling, see the [Situation Handling - Extended Framework](#) section.

The event specific details for Behavioral Insights are outlined in the table below:

Behavioral Insights event details	
Configuration Type	Details
Event Category	ABAP Class
Event Object	CL_PREDICT_RISK_SCORE
Event Type	RISK_FACTOR
Structure Name	I_BHVLINSGTSLTSTPRDTNRSLT

For more information on the authorization configuration for Situation Handling in SAP S/4HANA, see the [Define Authorizations for Apps \(Optional\)](#) section.

Troubleshooting

While running the prediction report, you might encounter some errors. The table below lists the error messages and the solutions for the errors:

Error Messages for Prediction Run Report		
Error Message	Description	Solution
Scenario is marked as inactive in Behavioral Insights	This error message is displayed when the prediction report is run for a scenario that is marked as inactive in Behavioral Insights.	Ensure that you choose a scenario that is active in Behavioral Insights, for running the prediction report. If the scenario is inactive, then access the Maintain Scenario customizing activity to activate it. For more information about maintaining a scenario, see the SAP Reference Customizing documentation.
The provided scenario does not exist in Intelligent Scenarios	This error message is displayed when the prediction report is run for a scenario that does not exist in ISLM.	Ensure that you select a scenario that has a corresponding predictive scenario created in ISLM and then mapped to the selected scenario in Behavioral Insight. You can use the Maintain Scenario customizing activity to assign an ISLM predictive scenario to a scenario in Behavioral Insights. For more information, see the SAP Reference Customizing documentation.
Scenario configuration is not maintained in Behavioral Insights	This error message is displayed when the prediction report is run for a scenario, which does not exist in Behavioral Insights.	Ensure that you select a Scenario ID, which has all the configuration information accurately maintained. You can configure scenario using the Maintain Scenario customizing activity. For more information about accurately

Error Message	Description	Solution
		configuring the scenario in Behavioral Insights, see the SAP Reference Customizing documentation.
The scenario configuration is incomplete in Intelligent Scenarios.	This error message is displayed when the prediction report is run for a scenario with incomplete information in ISLM.	Ensure that you verify the details for the scenario in ISLM. For more information, see Troubleshooting .
Not authorized to perform this action.	This error message is displayed when the user is not authorized to run prediction reports.	Ensure that you have access to run the prediction reports. You can contact your administrator for authorization privileges. For more information about authorization objects and privileges, see Security Guide for SAP S/4HANA 2020 .
The system did not find any records to match the selection criteria.	This error message is displayed when the prediction report is run for a scenario that does not contain data.	Ensure that there is adequate input data for the prediction scenario. Additionally, ensure that the configuration information for Behavioral Insights is accurately maintained using the Maintain Scenario customizing activity and also that customer data is accurately extracted from source tables using the Maintain Customer Extracts customizing activity. For more information, about the maintaining scenarios and maintaining customer extracts, see the SAP Reference Customizing documentation.

Explore At-Risk Customers

You can use the [Explore At-Risk Customers](#) app to analyze risk scores and trends across multiple scenarios, for example, Receivership, Late Payment, Late Filing, and Debtor Analysis.

The following scenarios are delivered by Behavioral Insights:

- Receivership: You can analyze the risk scores and trends for the customer that lead to receivership.
- Late Payment: You can analyze the risk scores and trends for the customer that lead to late payments.
- Late Filing: You can analyze the risk scores and trends for the customer that lead to late filings.
- Debtor Analysis: You can analyze the transactions and debt for the customer across all account types or tax types.
- Predicted Days to Payment: You can analyze the days-to-payment prediction for the tax paying entity.

You can also analyze the risk scores and risk trends. You can also study the customer pulse by viewing and analyzing the actions that they perform.

Filters

The [Explore At-Risk Customers](#) overview page offers a standard view that is pre-delivered by Behavioral Insights.

Visual Filter

The **Standard view** is displayed in the visual filter mode.

In the **Standard view**, the following details are displayed:

- **Number of Occurrences by Scenario:** The number of entities, customers, accounts or subaccounts, which are categorized by different scenarios. The default scenarios delivered are receivership, late filing, late payment, and debtor analysis.
- **Number of Occurrences by Risk Score Range:** The number of entities, customers, accounts or subaccounts, which are categorized using the high, medium or low risk score range for the selected scenario. The risk score range varies according to the scenario involved.

i Note

The risk score range filter is not applicable to the Debtor Analysis and Predicted Days to Payment scenarios.

- **Number of Occurrences by Debt Range:** The number of entities, customers, accounts or subaccounts, which are categorized using the high, medium or low debt range for the selected scenario. The debt range varies according to the scenario involved.

The risk score range and debt range selected according to the scenario define the additional details that are displayed for each customer in the **Table view**.

You can also click on each customer row in the **Table view** and navigate to the customer's details page, which displays the customer pulse, related risks details and event details for a particular entity.

The **Standard view** in the Explore At-Risk Customer app also provides additional filters and views to display the customer details for the scenarios.

Compact Filter

In addition to the **Visual filter** mode that is displayed as a default in the **Standard view**, you can also use the **Compact filter**, which provides the search functionality for viewing the customer details.

You can also use the following fields to filter and display the customer details for each scenario:

- **Scenario ID**
- **Risk Score Range**
- **Debt Range**
- **Customer ID**
- **Customer Type**
- **Account ID**
- **Subaccount ID**
- **Customer Status**
- **Predicted Days to Payment Range**

i Note

You must configure the **Predicted Days to Payment Range** in the **Maintain Predicted Days to Payment Range** activity. The IMG activities for this product are embedded within the IMG of your SAP application. You can use the transaction **SPRO** to access the IMG structure.

You can use the SAP Reference IMG documentation for information on managing all the IMG activities for the product.
The path for the **Behavioral Insights IMG** is:

► [SAP Customizing Implementation Guide](#) ► [Cross-Application Components](#) ► [Behavioral Insights](#) ►

i Note

- The fields in the **Compact Filter** mode can vary according to the scenario involved. For example, if you select the **Debtor Analysis** scenario, the **Risk Score Range**, **Account**, **Subaccount ID**, and the **Predicted Days to Payment** fields are not displayed.
- When you select the **Debtor Analysis** scenario, a single customer can have multiple account types. The debt range for the **Debtor Analysis** scenario is calculated for each customer account type, and the most critical debt range associated with the account type is displayed as the debt range for the customer.

Customizing Filters

Context

You can customize the fields on the Customer Details page.

Procedure

1. From the SAP Fiori App launchpad, select **Customer Analysis**, and then navigate to the **Explore At-Risk Customers** app.
2. Choose **Adapt Filters**. You can select either the **Visual Filter**, **Group View**, or **List View** for the filter visualization.
3. Choose the fields that you want to enable in the filter bar and click **OK**.
4. Click **Go**. To change the filter visualization, you can toggle between **Compact Filter** or **Visual Filter**.

Views

In the **Standard view** page, once you select the scenario, the system displays the corresponding scenario details in the **Chart View**, the **Table View**, or the **Chart and Table View**. You can use this information to analyze the customers risk score and risk trends.

Chart View

The chart view displays the information for entities who have risk scores in a chart form for the filtered selection.

Behavioral Insights provides a default heat map chart type to visualize the data. You can use the **View By** option to display the data based on the parameters you choose.

You can use the **Settings** option to select the parameters based on which the data is displayed in the **Chart View**. Changing the chart parameters makes different chart types available. The logical selection you make determines the chart type that you can use to display the customer information.

For example, the scatter plot chart requires at least 2 measures, so it would be available when you select the attributes **Debt**, **Risk Score**, and **Number of Occurrences**. The bubble chart requires at least 3 measures, so it would be available when you select the attributes **Risk Score**, **Number of Occurrences**, **Debt**, and **Credit**.

Table View

The table view displays the scenario risk scores for each entity in a tabular form. The table information displayed is based on the filtered selection or content area selection that you make in the chart. To view further details about the entity, you can click on the row or choose the details arrow in the final column of the table.

Some examples of fields that are displayed in the table view section according to the selected scenario are shown below:

- **Customer ID:** A unique identifier for the customer.
- **Customer Name:** The name of the customer.
- **Account ID:** The account ID of the customer.
- **Subaccount ID:** The subaccount ID of the customer.
- **Risk Score:** The risk score value follows a color scheme to indicate whether it belongs to the low, medium or high range. When you click the risk score, a dialog box appears. It provides information about the risk score and the top influencing factors. A red upward arrow indicates that the risk score value has increased, and a green downward arrow indicates that the risk score value has decreased.

For example, if the risk score is 90 from today's prediction run and was 80 from the previous prediction run, an up red arrow is displayed. If the risk score is 80 from today's prediction run and was 90 from the previous prediction run, then a down green arrow is displayed. If there is no difference in the risk, then no arrow is displayed.

i Note

The **Risk Score** field is not applicable for the **Predicted Days to Payment** scenario.

- **Risk Score Trend:** The risk score trend is displayed in the percentage value. When you click the risk score trend, a dialog box appears. It provides information about the risk score trend and a graph indicating the previous risk scores.

A red upward arrow indicates that the risk score trend has increased, and a green downward arrow indicates that the risk score trend has decreased.

i Note

The **Risk Score** field is not applicable for the **Predicted Days to Payment** scenario.

- **Debt:** The debt owed by the entity in the local currency.
- **All High Risks:** The number of high-risk occurrences for the customer. When you choose the risk number, a dialog box appears. It displays the scenario name with the risk number.
- **Predicted Days to Payment Range:** The range for the days to payment prediction.

i Note

The **Predicted Days to Payment Range** field is not applicable for the other scenarios.

You can use the **Settings** option to personalize the table fields.

You can use the **Show Details** option to view additional details for the customer, such as:

- **Account Name:** The account name of the customer.
- **All Medium Risks:** The number of medium-risk occurrences for the customer. When you choose the risk number, a dialog box appears. It displays the scenario name with the risk number.

- **All Low Risks:** The number of low-risk occurrences for the customer. When you choose the risk number, a dialog box appears. It displays the scenario name with the risk number.

View Information

The following actions can be performed in the **Chart view** and/or the **Table view**.

Action	Description
Details	You can select the chart area to view the entity information details plotted on the chart.
View By	You can use the View By option to select the entity attributes that you want to add to the chart. The selected attributes are displayed along the x-axis of the chart.
Toggle Legend Visibility	You can use the Toggle Legend Visibility option to zoom the chart in or out to increase its visibility.
Settings	You can define the view settings for the chart and table and select the entity attributes that must be visible in the chart or table.
Maximize or Minimize	You can use the Maximize or Minimize option to either maximize or minimize the display of the chart or table.
Chart and Table View	You can select the Chart and Table View to view both the chart and table for the filtered selection.
Chart View	You can select the Chart View to only view the chart for the filtered selection.
Table View	You can select the Table View to only view the table for the filtered selection.
Show All Items	You can select Show All Items to display all items based on the filtered selection.
Show Only Chart Items	You can select Show Only Chart Items to only display items based on the chart selection.
Export to Spreadsheet	You can use the Export to Spreadsheet option to download the list of selected entities in a .csv format file in the table view.

Customer Details

The **Customer Details** page displays the table details in the header area of the overview page, and provides three tabs **Customer Pulse**, **Related Risks** and **Events** to visualize and analyze the entity's event data in both graphical and table formats.

Once you have selected the details arrow next to an entity in the table view of the overview page, the **Customer Details** page appears.

The customer details displayed in the header area vary according to the scenario selected. The following information is displayed in the header:

- **Customer ID**

- [Account Name](#)
- [Customer Type](#)
- [Customer Status](#)
- [Scenario](#)
- [Subaccount ID](#)
- [Risk Score Trend](#)
- [Risk Score Range](#)
- [Risk Score](#)
- [Debt](#)
- [Debt Range](#)
- [Credit](#)
- [Predicted Days to Payment Range](#)

i Note

When you are viewing the customer details for the [Debtor Analysis](#) scenario, you can click on the cumulative debt displayed in the [Debt](#) field. The customer debt for each account category is then displayed on a dialog box.

Customer Pulse

The customer pulse is a graphical representation of the customer events based on the time at which they occurred and the total debt at that point in time.

Once you select a time period, the events that have occurred during that time period are available in the filter selection [All](#). Additional filters, such as [Contract Account Category](#), [Subaccount ID](#), [Event Category](#) and [Period Key](#) are available according to the scenario involved, which you can select to visualize in the graph.

i Note

In the [Contract Account Category](#) filter, you can select single or multiple account categories to filter and view the details in the [Customer Pulse](#).

The graph plots the debt on the y-axis and the time period on the x-axis. The coordinates plotted in the graph are the events that occurred for that time period and the total debt at that point in time.

i Note

Not every coordinate plotted on the graph is connected. A relative trend line is plotted in the graph.

When you click or hover over the plotted coordinates, the following information is displayed on a dialog box:

- [Event Start Date](#)
- [Event Name](#)
- [Account Name](#)
- [Subaccount ID](#)

- [Actual Amount](#)
- [Debt](#)
- [Period Key](#)

You can choose [Show Icons](#) or [Hide Icons](#) to either display or hide the icons in the graph.

i Note

The icons are displayed as a default if the selected time period is less than two years. Any influencing events that have occurred during the selected time period are highlighted in blue.

Based on the selected time period, the events that have occurred during that time period are plotted in the [Customer Pulse](#) chart. You can also use the additional filters, such as [Account ID](#), [Subaccount ID](#), [Event Category](#) and [Period Key](#) that are available on the smart filter tab to further visualize the filtered list of events in the graph.

By default, data is displayed for 6 months if no influencing factors are available. If the influencing factors are available, the time period is selected based on the reasons configured in the [Maintain Reasons](#) activity. For more information about accessing the Customizing information, see the **Maintain Configuration Information** section in [Quick Reference](#).

The Customizing activities for this product are embedded within Customizing for your SAP application. You can use the transaction *SPRO* to access the Customizing structure.

You can refer to the SAP Reference IMG documentation for information about managing all Customizing activities for the product. The Behavioral Insights Customizing activities are accessed as follows:

SAP Customizing Implementation Guide > Cross-Application Components > Behavioral Insights

i Note

The [Account ID](#) and [Subaccount ID](#) filters are displayed according to the level of the scenario selected.

- If the scenario selected is found at customer level, both the [Account ID](#) and [Subaccount ID](#) filters are displayed
- If the scenario selected is found at account level, the [Account ID](#) filter is displayed.
- If the scenario selected is found at subaccount level, neither the [Account ID](#) nor the [Subaccount ID](#) filters are displayed.

In the graph area, you can zoom out to expand the time period and view the details. The following lines are plotted in the graph:

- **Debt lines:** The debt lines are plotted to indicate the category in which the debt value is available for each event. The category is based on the debt range configured in the **Maintain Range** configuration activity. For more information about accessing the Customizing information, see the **Maintain Configuration Information** section in [Quick Reference](#).

The lines fall between [Medium Debt](#) and [High Debt](#), with the semantic colouring based on the debt range category values [Low](#), [Medium](#), and [High](#).

Related Risks

Any other scenario risks that are related to the customer, including risks associated with other accounts or subaccounts for the same customer are displayed in this section. You can click the row or the details arrow in the [Related Risks](#) section to view further details.

i Note

You can personalize the fields in [Related Risks](#) table using the [Settings](#) option. For example, to view additional details while using the [Predicted Days to Payment](#) scenario, you can add the [Predicted Days to Payment](#) field to view the prediction in the actual number of days, in addition to the range.

Events

The events related to a customer are displayed in a table format in the [Events](#) section.

i Note

You can personalize the fields in the events table using the [Settings](#) option.

The following table displays some examples of the event table fields:

Field	Description
Event Name	Name of the event.
Event Category	The event names are the subsets of an event category. For example, Payment is the corresponding event category for the event names Full Payment and Partial Payment.
Event Value	The value associated with the event. For example, the event category status can have values such as Live and Strike-off .
Actual Amount	The actual amount for that particular event. For example, if the event is payment, the actual is the amount that the customer needs to pay. Whereas, if the event is filing, the actual is the amount to be paid to the customer.
Debt	Amount that is due when the event occurred.
Event Start Date	Start date of the event.
Account ID	Account ID of the customer.
Subaccount ID	Subaccount ID of the customer.
Account Name	Account name of the customer.
Credit	Amount that has been paid by the customer.

Standard View

The At-Risk Customers Overview page offers a Standard View that is pre-delivered by Behavioral Insight

In the Standard View, the following filters are displayed:

- Scenario
- Risk Score Range
- Debt Range

- Collection Strategy
- Collection Step
- Risk Reason
- Date Range

In the Standard View, the following charts are displayed:

- Customers at Risk
- Debt Amount
- Top 5 Risk Reasons
- Collection Strategy Trend
- Top 5 Collection Steps Trend
- Collection Analysis

At-Risk Customers Overview

You can use the [At-Risk Customers Overview](#) app to view trends like collection strategy, customer groups, and more, which are represented visually, across multiple scenarios like Receivership, Late Payment, and Late Filing.

Filters

The following filters are available for you to choose from in the [Smart Filter](#) bar:

- [Scenario](#)
- [Risk Score Range](#)
- [Debt Range](#)
- [Collection Strategy](#)
- [Collection Step](#)
- [Risk Reason](#)
- [Date Range](#)

i Note

You can use the [Adapt Filters](#) option to [Show](#) or [Hide](#) the standard filters on the [Smart Filter](#) bar.

Charts

The following charts are visualised based on the options you choose in the [Smart Filter](#) bar:

- [Customers at Risk](#): The [Customers at Risk](#) chart displays the total number of customers who are at risk for the selected scenario, by [Collection Strategy](#), [Risk Score Range](#), or [Debt Range](#).

- **Total Debt Amount:** The **Total Debt Amount** chart displays the total debt amount of the customers by **Collection Strategy**, **Risk Score Range**, or **Debt Range**.
- **Top 5 Risk Reasons:** The **Top 5 Risk Reasons** chart displays the top five risk reasons by **Strong Confidence** or **Strong and Moderate Confidence**.
- **Collection Strategy Trend:** The **Collection Strategy Trend** chart displays the average debt amount or the average risk score trend by time periods.
- **Top 5 Collection Steps Trend:** The **Top 5 Collection Steps Trend** chart displays the top five collection steps trend by average risk score or average debt amount.
- **Collection Analysis:** The **Collection Analysis** chart displays the average risk score by average debt amount details by collection steps or collection strategy.
- **Customer Groups:** The **Customer Groups** chart displays the number of customers by debt amount. When you click on the indicators on the chart, the **Customer Group** pop-up displays the deviation from the top grouping indicators such as **Average Due Amount** and **Average Months Since Registration**. You can click on **View all Customers** to view the customer details in the **Explore At-Risk Customers** application.

i Note

The information displayed in the **Customer Groups** chart is derived from machine learning clustering algorithms. Currently, this information is available only for the **Late Payment** scenario.

i Note

You can click on the individual chart title tab to navigate to the customer details in the **Explore At-Risk Customers** application.

Analytics Using SAP Analytics Cloud

SAP Analytics Cloud (SAC) is a business intelligence platform that provides analytics capabilities for visualizing live data.

You can use SAC to create models from the data sources, build stories based on those models, and perform live analysis using the CDS views delivered by Behavioral Insights.

Behavioral Insights delivers the following views:

- Prediction History
- Customer Events History
- Prediction Result Reason

To use SAC to perform analytics, you must:

1. [Build a connection to the S/4HANA system](#)
2. [Build a model for the scenario](#)
3. [Create a story to analyze the data model](#)

Build a Connection to the S/4HANA System

You can create live data connections to the S/4HANA system. The data is live, meaning that when a user opens a story in SAP Analytics Cloud, any changes made to the data in the source system are reflected immediately.

For detailed information about building a direction connection to the S/4HANA system, see [Live Data Connection to SAP S/4HANA On-Premise Using a Direct CORS Connection via Unified Connectivity](#).

For additional information on building live connections to S/4HANA system, see [Live Data Connections to SAP S/4HANA](#).

Build a Model for a Scenario

You can use the live connection you created in the S/4HANA system to build a model for a scenario.

For detailed information about building a model for a scenario, see [Creating a Model from a Live Data Connection](#).

While building the model, you can use one of the following CDS views delivered by Behavioral Insights as the data source:

- 2CCBIPRDTNHISTQ - Prediction History
- 2CCBICUSTEVTHISTQ - Customer Events History Query
- 2CCBIPRDTNRSLTRSNQ - Prediction Result Reason

Create a Story to Analyze the Data Model

You can use a story to describe your data that also includes charts, tables, text, images. You can use the data from one or more models that you have created to create a story and analyze the data effectively.

For detailed information about creating a story to analyze the data model, see [Creating a New Story](#).

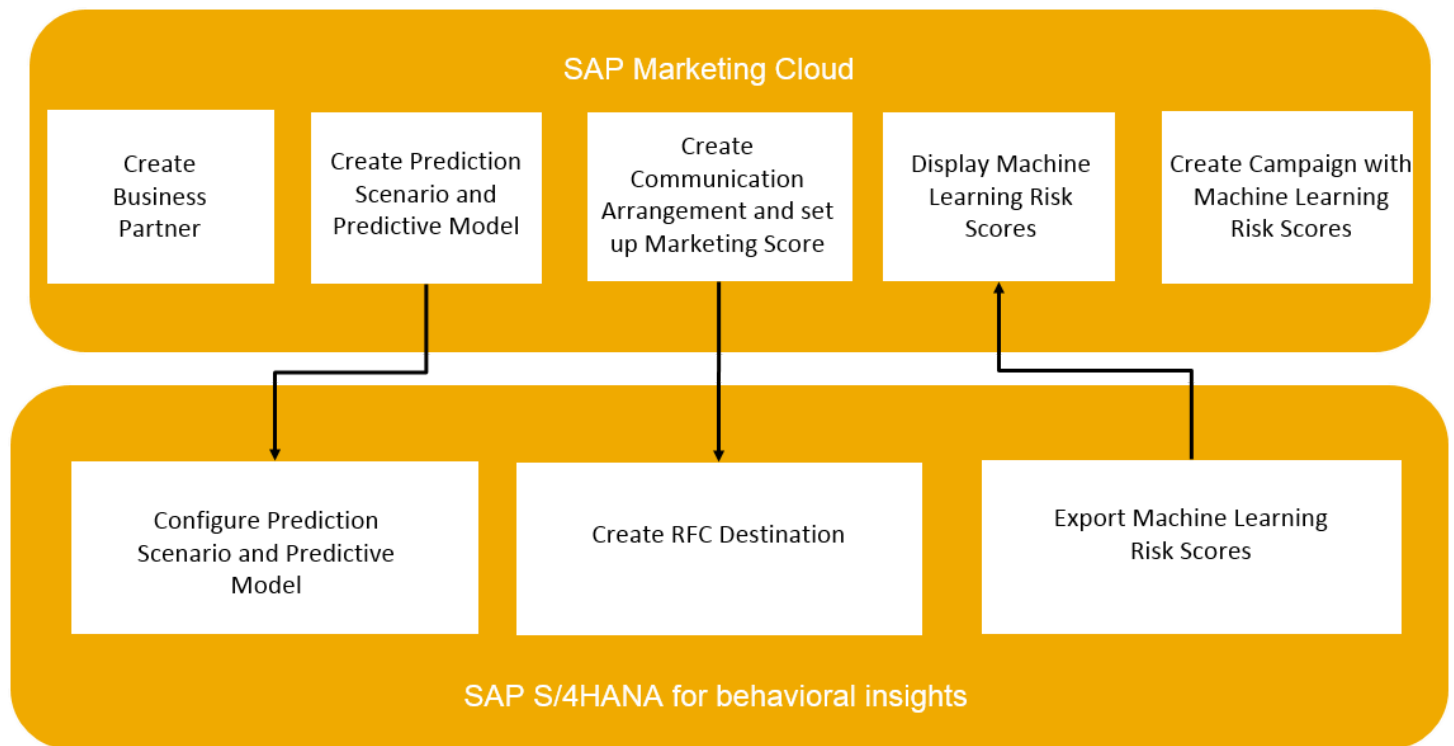
While creating the story, you can select the model created using the Behavioral Insights CDS views to analyse the prediction data.

Integration with SAP Marketing Cloud

Behavioral Insights offers standard integration with SAP Marketing Cloud. Machine Learning risk scores can be sent to SAP Marketing Cloud for further actions.

The marketing score values for the corresponding customers can be updated in SAP Marketing Cloud. This information can be used for messaging purposes. For more information, see the documentation on [SAP Marketing Cloud](#).

Integration with SAP Marketing Cloud Workflow



Pre-requisites

Before integrating Behavioral Insights with SAP Marketing Cloud, you must have completed the following procedures:

- **Create a business partner in SAP Marketing Cloud:** You can use the information detailed in the Create Contacts with Additional IDs section in [Payload Examples](#) to create the business partner.
- **Create a prediction scenario and predictive model for the Behavioral Insights machine learning scenarios in SAP Marketing Cloud:** Once the business partner is created in SAP Marketing Cloud, you must then create a prediction scenario and predictive model for all the three Behavioral Insights machine learning scenarios - Receivership, Late Filing and Late Payment, in SAP Marketing Cloud.

The following roles are required to create the prediction scenario and predictive model:

- For **Predictive Scenarios** app - *ADMINISTRATOR_MKT*
- For **Predictive Studio** app - *BUSINESS_ANALYST_MKT*

You can use the information detailed in the [Uploading External Score Values in Predictive Studio](#) to create a prediction scenario and a predictive model.

- **Create communication arrangements and marketing scores in SAP Marketing Cloud:** Once you configure the prediction scenarios and the prediction models in Behavioral Insights, you must then create the communication arrangements and marketing scores in SAP Marketing Cloud.

The Communication Arrangement allows you to integrate your system or solution with other systems to enable data exchange. You can use the information detailed in [Communication Management](#) to set up a communication arrangement. You must use the *SAP_COM_0307* communication scenario to set up the communication arrangement.

You must also use the *API_MKT_SCORE_SRV* to set up the marketing scores. For more information, see [Scores](#).

Configuration Information

You must complete the following configuration procedures:

- **Configuring prediction scenarios and prediction models in Behavioral Insights**

You must configure the prediction scenarios in Behavioral Insights. You must perform the procedure for each of the three machine learning scenarios - Receivership, Late Filing, and Late Payment.

i Note

Currently the Debtor Analysis scenario cannot be configured for analysis in SAP Marketing cloud.

You must configure the [External Predictive Scenario ID](#) and the [External Model ID](#) in the [Maintain Scenario](#) activity. For more information about accessing the customizing information, see the [Maintain Configuration Information](#) section in [Quick Reference](#).

The IMG activities for this product are embedded within the IMG of your SAP application. You can use the transaction *SPRO* to access the IMG structure.

You can use the SAP Reference IMG documentation for information on managing all the IMG activities for the product. The path for the Behavioral Insights IMG is:

► [SAP Customizing Implementation Guide](#) ► [Cross-Application Components](#) ► [Behavioral Insights](#) ►

- **Creating RFC destination**

After creating the Communication Arrangement and setting up the scores in SAP Marketing Cloud, you must specify the details of the communication arrangement in the RFC destination for Behavioral Insights using the *SM59*.

i Note

Ensure that the RFC destination name is *BEI_SMC_HTTP*.

Exporting Machine Learning Risk Scores to SAP Marketing Cloud

Behavioral Insights provides you with the following job templates to export the risk scores to SAP Marketing Cloud:

- **Marketing Scores Export Report Template for Behavioral Insights:** You can use this template to export the marketing scores for each scenario.
- **Risk Prediction with Marketing Scores Export Report Template for Behavioral Insights:** You can use this template to sequentially run the prediction report and then export the marketing scores, for each scenario.

You can use the [Maintain Application Jobs](#) application to run the job templates. For a more detailed overview of the standard key features offered by this app, see [Maintain Application Jobs](#).

i Note

- Currently, you can only run the above job templates for the following three machine learning scenarios - Receivership, Late Payment and Late Filing.
- While running the above job templates, you must select a [Range ID](#) to filter the data for the scenarios based on the [Range ID](#) selected.

APIs for Behavioral Insights

The following table contains all APIs available in Behavioral Insights.

i Note

The API documentation is available in English only.

Overview of APIs in Behavioral Insights

Name	Technical Name	Type	Application Component
External Events - Read, Create (A2X)	API_EXTERNALEVENTS	OData	Behavioral Insights

External Events - Read, Create (A2X)

Technical name: API_EXTERNALEVENTS

This service enables you to read or create Behavioral Insights External Event data with the data provided in a payload, in an API call. It is based on the OData protocol, and can be consumed in SAP Fiori apps and on other user interfaces.

This service is published on the SAP Business Accelerator Hub. For more information about APIs, see [APIs on SAP Business Accelerator Hub](#).

Service Structure

The service is built using the CDS views based on Behavioral Insights External Events table.

This service supports standard system query options for the OData protocol, such as \$filter, \$top, \$orderby, \$skip, \$inlinecount, \$select, and \$expand.

Service Header

The service message header contains information on the service, the involved sender and receiver as well as date and time.

Entity

The entity contain the business data of the service.

Entity	Description	Link to Details
A_BhvrInsngtsExternalEvent	Behavioral Insights External Events	External Events

i Note

To display more information about the properties on the SAP Business Accelerator Hub, open one of the entity's operations and select **Model** or **Schema**.

Activate and Maintain Service

The transaction for activating and maintaining services is used to maintain all registered services on the SAP Gateway server (hub system) and to register and activate services. For more information, see [Activate and Maintain Services](#).

Once the service is activated, you must create a user role to be able to access this service. For more information, see [Roles in the SAP Gateway Landscape](#).

Error Handling

The error log facilitates easy navigation to the affected source code where you can make the necessary corrections. For more information, see [Error Log](#).

External Events

Parameters

Parameter	Description	Necessity	Format
<i>CustomerEventRecordUUID</i>	Universally unique identifier for the customer event record. Behavioral Insights creates a unique ID if the customer does not provide a unique customer event record ID.	Optional	<code>string(\$uuid)</code> Example: 01234567-89ab-cdef-0123-456789abcdef
<i>CustomerEventRecordID</i>	Unique identifier for the customer event record	Optional	<code>string</code> Example:0123456789abcdef0123456789abcdef
<i>BusinessPartner</i>	Identifier for the business partner. The Business Partner ID provided should exist in SAP S/4HANA.	Mandatory	<code>String</code>
<i>BhvrlInsgtsEventNumber</i>	Behavioral Insights event number. The customer must provide either the Behavioral Insights event number or the External Event ID to create external event data in Behavioral Insights.	Optional	<code>integer(\$int16)</code>
<i>ExternalEventID</i>	Behavioral Insights External Event identifier. A unique ID for an event from an external system. The External Event ID must have been previously configured using the Maintain Events customizing activity. For more information about accessing the customizing information, see the Maintain Events Information section in Quick Reference . The IMG activities for this product are embedded within the IMG of your SAP application. You can use the transaction SPRO to access the IMG structure.	Optional	<code>string</code>

Parameter	Description	Necessity	Format
	You can use the SAP Reference IMG documentation for information on managing all the IMG activities for the product. The path for the Behavioral Insights IMG is: SAP Customizing Implementation Guide > Cross-Application Components > Behavioral Insights		
<i>CustomerEventStartDateTime</i>	Timestamp that indicates the start time and date of the customer event.	Optional	String Example: /Date(1492098664000)/
<i>CustomerEventEndDateTime</i>	Timestamp that indicates the end time and date of the customer event.	Optional	String Example: /Date(1492098664000)/
<i>OpenAmountInLocalCurrency</i>	Pending amount to be paid by the business partner indicated in local currency.	Optional	string(\$decimal)
<i>ClearedAmountInLocalCurrency</i>	Amount paid by the business partner indicated in local currency.	Optional	string(\$decimal)
<i>LocalCurrency</i>	Local currency of the business partner.	Optional	string
<i>BhvrInsgtsPeriodCode</i>	Period Code for the Behavioral Insights event.	Optional	string
<i>CustomerEventValue</i>	Value for customer event.	Optional	string
<i>CustomerEventValueTxt</i>	Value text for customer event.	Optional	string
<i>BhvrInsgtsAccountNumber</i>	Account number of the business partner.	Optional	string
<i>BhvrInsgtsAccountCategory</i>	Account category of the business partner.	Optional	string
<i>BhvrInsgtsSubAccountNumber</i>	Subaccount number of the business partner.	Optional	string